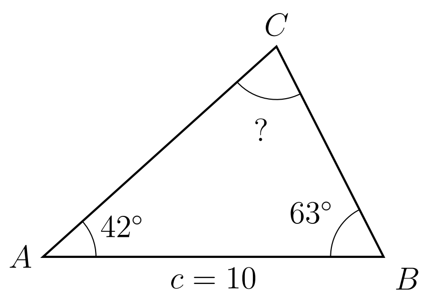
Solving AAS and ASA Triangles with the Law of Sines

Finding the third angle, solving AAS and ASA triangles, and using the law of sines in surveying problems

Directions.

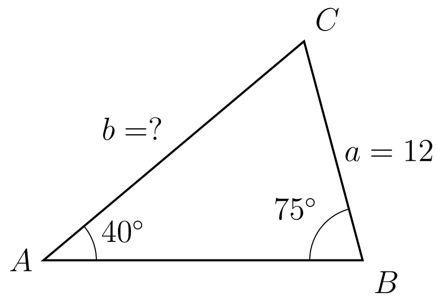
- Every triangle is drawn to scale from its own given values. Side a lies opposite angle A, side b opposite B, side c opposite C; the unknown you are asked for is marked with a question mark.
- Law of sines: $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$. Pair each side with the angle *across* from it.
- If the side you know and the side you want are not both paired with a known angle, find the third angle first.
- Round side lengths to two decimal places; angles are exact here.

1. In this triangle $A = 42°$, $B = 63°$ and $c = 10$. Find angle C.



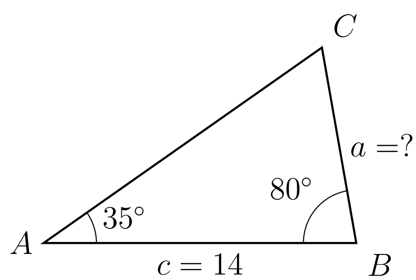
Answer: ________________ degrees

2. Here $A = 40°$, $B = 75°$ and $a = 12$. Find b.



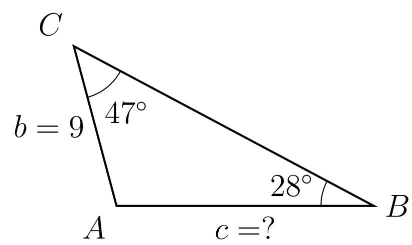
Answer: ____________________

3. Here $A = 35°$, $B = 80°$ and $c = 14$. Find a.



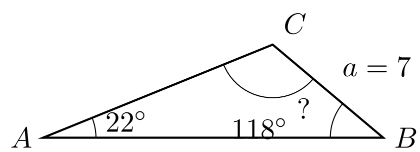
Answer: ____________________

4. Here $B = 28°$, $C = 47°$ and $b = 9$. Find c.



Answer: ________________

5. In this triangle $A = 22°$, $C = 118°$ and $a = 7$.



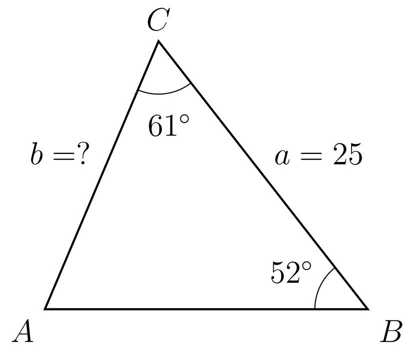
(a) Find angle B.

Answer: ________________ degrees

(b) Two angles and one side were enough to pin down this triangle completely. Explain why.

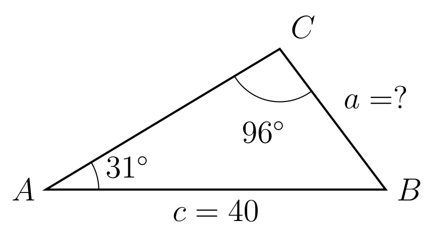
Answer: ________________

6. Here $B = 52°$, $C = 61°$ and $a = 25$. Find b.



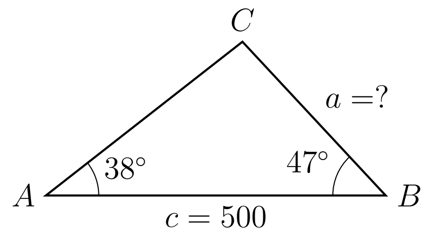
Answer: ______________

7. Here $A = 31°$, $C = 96°$ and $c = 40$. Find a.



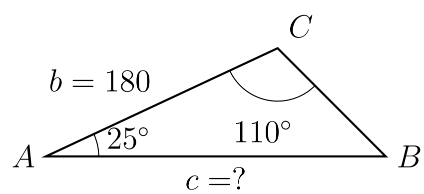
Answer: ______________

8. Two surveyors stand at points A and B on a straight road, $c = 500$ metres apart. Both sight the same radio mast at C. The angle at A, between the road and the line of sight, is $38°$; the angle at B is $47°$. Find a, the distance from B to the mast, in metres.



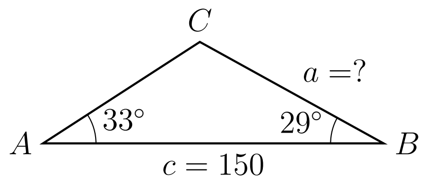
Answer: ________________ m

9. A straight footpath of length $b = 180$ metres joins a bench at A to a fountain at C. A statue stands at B. At the bench the angle between the path and the line to the statue is $25°$; at the fountain that angle is $110°$. Find c, the distance from the bench to the statue, in metres.



Answer: ________________ m

10. Two fire lookout towers stand at A and B, a distance $c = 150$ metres apart. Both spot the same column of smoke at C. The angle at tower A is $33°$ and the angle at tower B is $29°$.



(a) Find a, the distance from tower B to the smoke, in metres.

Answer: ________________ m

(b) Explain why this data determines exactly one triangle, while knowing two sides and an angle that is not between them may not.

Answer: ________________